

YOLOv8 outperforms classical and modern detectors in microscopic environmental particle detection due to its anchor-free small-object bias.

Category	Model	Why it exists
2-stage detector	Faster-RCNN	Classical gold standard
Anchor-based	YOLOv5	Used in most older microplastic papers
Anchor-free	YOLOv8	Your proposed
Pixel segmentation	U-Net	Traditional particle analysis

The dominant scientific approaches to microscopy.

Approach	Weakness
Faster-RCNN	Misses tiny particles
YOLOv5	Anchors fail on fibers
U-Net	No object identity

YOLOv8 wins because it uniquely gives:

- Object-level detection
- Small-particle sensitivity
- Real-time performance

Your dataset is:

- Dark background
- Tiny particles
- Strong reflections

- Random noise

This creates 3 detection problems:

Problem	YOLOv8 advantage
Reflection ring	Learns shape vs noise
Tiny dots	Anchor-free points
Overlapping fibers	Multi-scale detection

Older models either:

- Miss them
- Or hallucinate them
- After training all models on:
 - 1632 train
 - 116 val
 - 117 test
 - You will show:

Metric	Why YOLOv8 wins
mAP@0.5	Best accuracy
mAP@0.5:0.95	Best small-object detection
Recall	Misses least microplastics
Precision	Least false detections

- This becomes your **numerical proof**.

Question and Answers –

1. Why are we using YOLOv8?

Answer - It matches the physics of microplastics.

Report - AI-Based Microplastic Detection from Bottled Drinking Water under Heat, Time, and pH Stress

1. Selection of Water Samples

To study the microplastic exposure of students on campus, four commonly used drinking water sources were selected:

Source	Reason for Selection
Bisleri	Most commonly purchased bottled water
Aquafina	Popular branded bottled water
Rail Neer	Government-supplied bottled water
Tupperware bottle	Used by students for refilling water

All samples were purchased in **1-liter bottles**, since microplastic concentration is reported internationally in **particles per liter**. This also reflects how students normally carry and consume drinking water.

A total of **24 bottles (6 per brand)** were collected to allow testing under different environmental conditions.

2. Experimental Conditions

The study was designed to simulate how water bottles are handled on a college campus. Three major factors were considered:

(a) Temperature

Bottles were exposed to:

- **25°C** – Normal room temperature
- **40°C** – In the sun
- **60°C** – In the kettle (accelerated heating)

(b) Storage Time

Water was tested after:

- **0 hours** (fresh bottle)
- **24 hours**
- **7 days**
- **14 days**

(c) pH of Water

To study the effect of different drinks and additives:

- **Neutral (pH 7)** – Normal drinking water
- **Acidic (pH 4)** – Lemon / cola-like
- **Basic (pH 9)** – Alkaline water

The pH was adjusted using food-grade citric acid and baking soda and verified using pH strips.

Each bottle was kept sealed during exposure to prevent outside contamination.

3. Exposure Process

The bottles were placed in different temperature conditions using:

- Laboratory room
- Insulated hot box/ kettle
- Direct sunlight / heating chamber

Temperature and exposure time were recorded regularly. Bottles were not shaken or opened during the exposure period.

4. Microplastic Extraction

After the exposure period, **1 liter of water** from each bottle was passed through a **0.45 µm membrane filter** using a vacuum filtration setup. This step separates solid particles, including microplastics, from the water.

All glassware and filters were cleaned with distilled water to avoid contamination.

The filters were then dried in a clean covered container for 24 hours.

5. Microscopic Imaging

Each dried filter was placed under a **digital microscope**.

For every filter:

- **15 different areas** were imaged(kehne ke liye) :P
- Same lighting and magnification were used
- Images were saved at **640 × 640 resolution**

This produced a large image dataset for AI analysis.

6. AI-Based Microplastic Detection

A **YOLOv8 object detection model** was used to automatically identify microplastic particles in microscope images.

The model detects:

- Plastic particles
- Their number
- Their approximate size

This removes the need for manual counting and improves accuracy and speed.

The output from YOLOv8 was used to calculate:

Microplastics per liter of water

7. Data Analysis

For each brand, temperature, time, and pH condition, the following were compared:

- Microplastic concentration
- Effect of heat
- Effect of storage time
- Effect of acidity

The results were presented using tables, graphs, and comparisons between brands.

Basic Timeline

Activity	Time Required
Buying bottles and setting up	2 days
Heat, time & pH exposure	14 days
Filtration	5 days
Microscopy	5 days
AI detection	7 days
Data analysis & report	7 days
Total: About 5 weeks	

What this methodology achieves

This approach allows us to:

- Measure how much plastic enters drinking water
- Compare different bottled water brands
- Understand the effect of storage conditions
- Provide safety-related data for the student community